2.21 Stepper motor

Stepper motor

Overview

In this lesson, you will learn a fun and easy way to drive a stepper motor.
The stepper we are using comes with its own driver board making it easy to connect to our ESP32.

Component Required:

(1) x Elegoo ESP32

(2) x 400 tie-points breadboard

(1) x ULN2003 stepper motor driver module

(1) x Stepper motor

(1) x 9V 1A Adapter

(1) x Power supply module

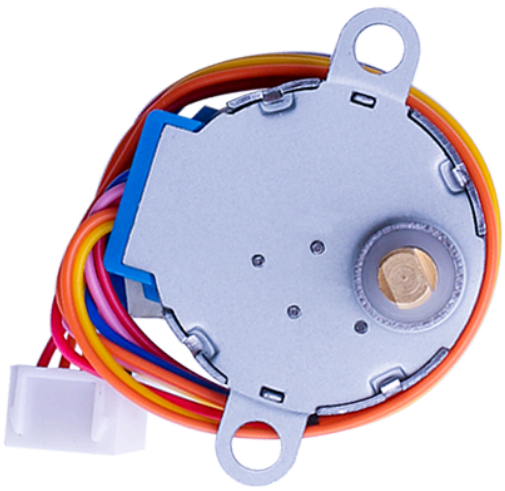
(6) x F-M wires (Female to Male DuPont wires)

(2) x M-M wire (Male to Male jumper wire)

Component Introduction

Stepper Motor

A stepper motor is an electromechanical device which converts electrical pulses into discrete mechanical movements. The shaft or spindle of a stepper motor rotates in discrete step increments when electrical command pulses are applied to it in the proper sequence.
The motors rotation has several direct relationships to these applied input pulses. The sequence of the applied pulses is directly related to the direction of motor shafts rotation.
The speed of the motor shafts rotation is directly related to the frequency of the input pulses and the length of rotation is directly related to the number of input pulses applied.
One of the most significant advantages of a stepper motor is its ability to be accurately controlled in an open loop system. Open loop control means no feedback information about position is needed. This type of control eliminates the need for expensive sensing and feedback devices such as optical encoders.
Your position is known simply by keeping track of the input step pulses.



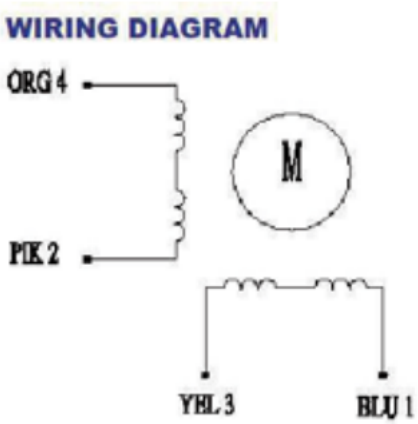
Stepper motor 28BYJ-48 Parameters

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Parameter	Value/Specification	Parameter	Value/Specification
Model	28BYJ-48	In-traction Torque	>34.3mN.m(120Hz)
Rated voltage	5VDC	Self-positioning Torque	>34.3mN.m
Number of Phase	4	Friction torque	600-1200 gf.cm
Speed Variation Ratio	1/64	Pull in torque	300 gf.cm
Stride Angle	5.625°/64	Insulated resistance	>10MΩ(500V)
Frequency	100Hz	Insulated electricity power	600VAC/1mA/1s
DC resistance	50Ω±7%(25℃)	Insulation grade	A
Idle In-traction Frequency	> 600Hz	Rise in Temperature	<40K(120Hz)
Idle Out-traction Frequency	> 1000Hz	Noise	<35dB(120Hz,No load,10cm)

Interfacing circuits

The bipolar stepper motor usually has four wires coming out of it. Unlike unipolar steppers, bipolar steppers have no common center connection. They have two independent sets of coils instead. You can distinguish them from unipolar steppers by measuring the resistance between the wires. You should find two pairs of wires with equal resistance. If you've got the leads of your meter connected to two wires that are not connected (i.e. not attached to the same coil), you should see infinite resistance (or no continuity).



ULN2003 Driver Board

·Size: 42mmx30mm

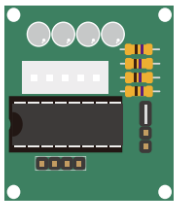
·Use ULN2003 driver chip, 500mA

·A. B. C. D LED indicating the four phase stepper motor working condition.

·White jack is the four phase stepper motor standard jack.

·Power pins are separated

·We kept the rest pins of the ULN2003 chip for your further prototyping.

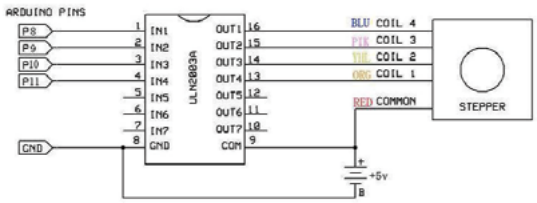


ULN2003 Driver Board

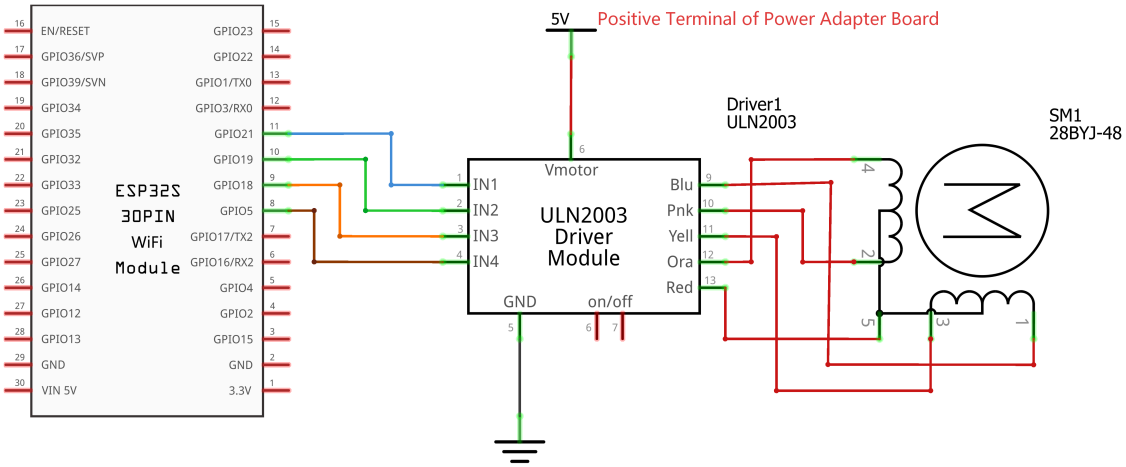
The simplest way of interfacing a unipolar stepper to Arduino is to use a breakout for ULN2003A transistor array chip. The ULN2003A contains seven Darlington transistor drivers and is somewhat like having seven TIP120 transistors all in one package. The ULN2003A can pass up to 500 mA per channel and has an internal voltage drop of about 1V when on. It also contains internal clamp diodes to dissipate voltage spikes when driving inductive loads. To control the stepper, apply voltage to each of the coils in a specific sequence.
The sequence would go like this:

LeadWire Color	--> CW Direction (1-2 Phase) Back to Elegoo							
	1	2	3	4	5	6	7	8
4 ORG	-	-						
3 YEL			◇	-				
2 PIK				-	-			
1 BLU						-	-	-

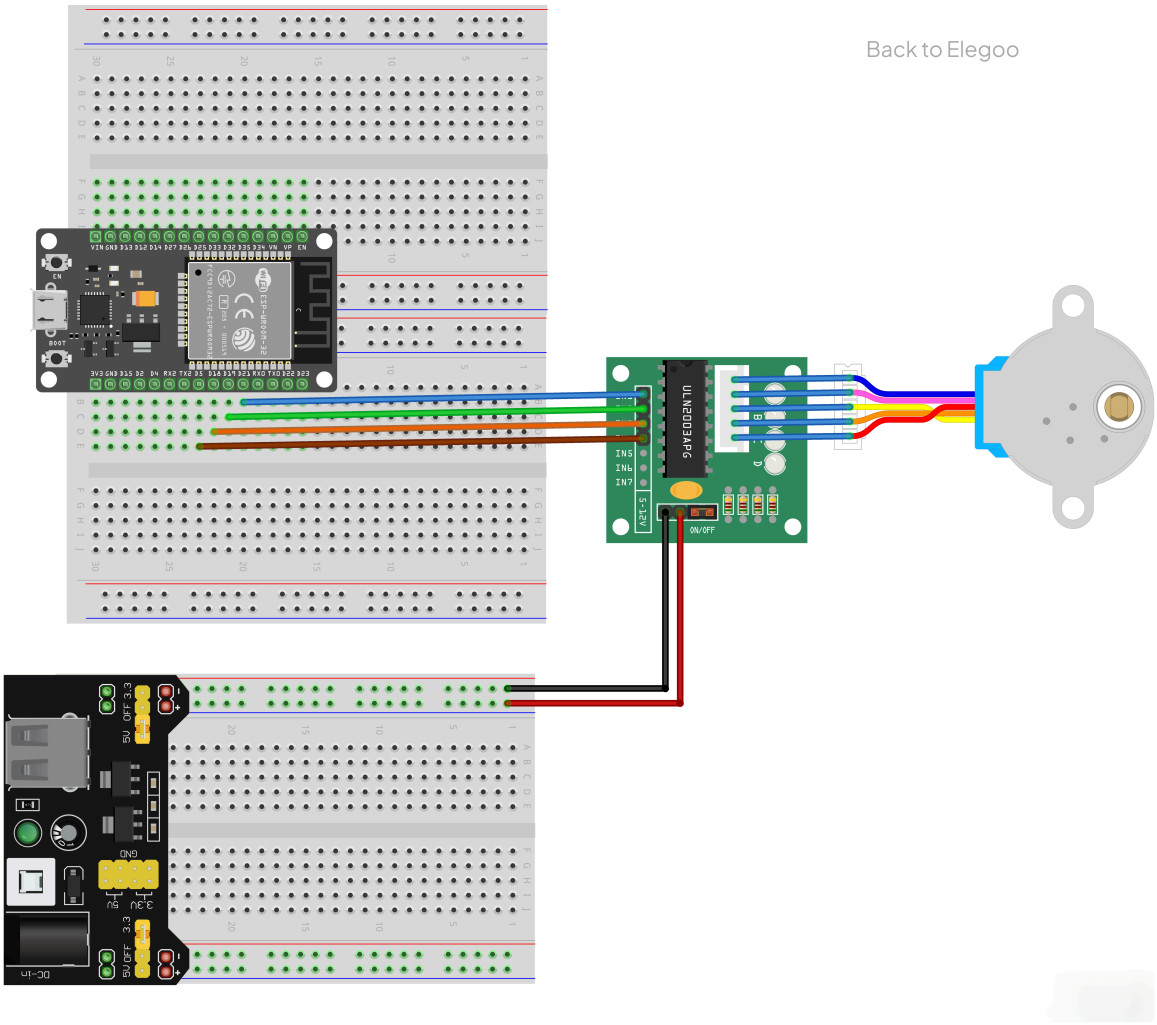
Here are schematics showing how to interface a unipolar stepper motor to four controller pins using a ULN2003A, and showing how to interface using four com.



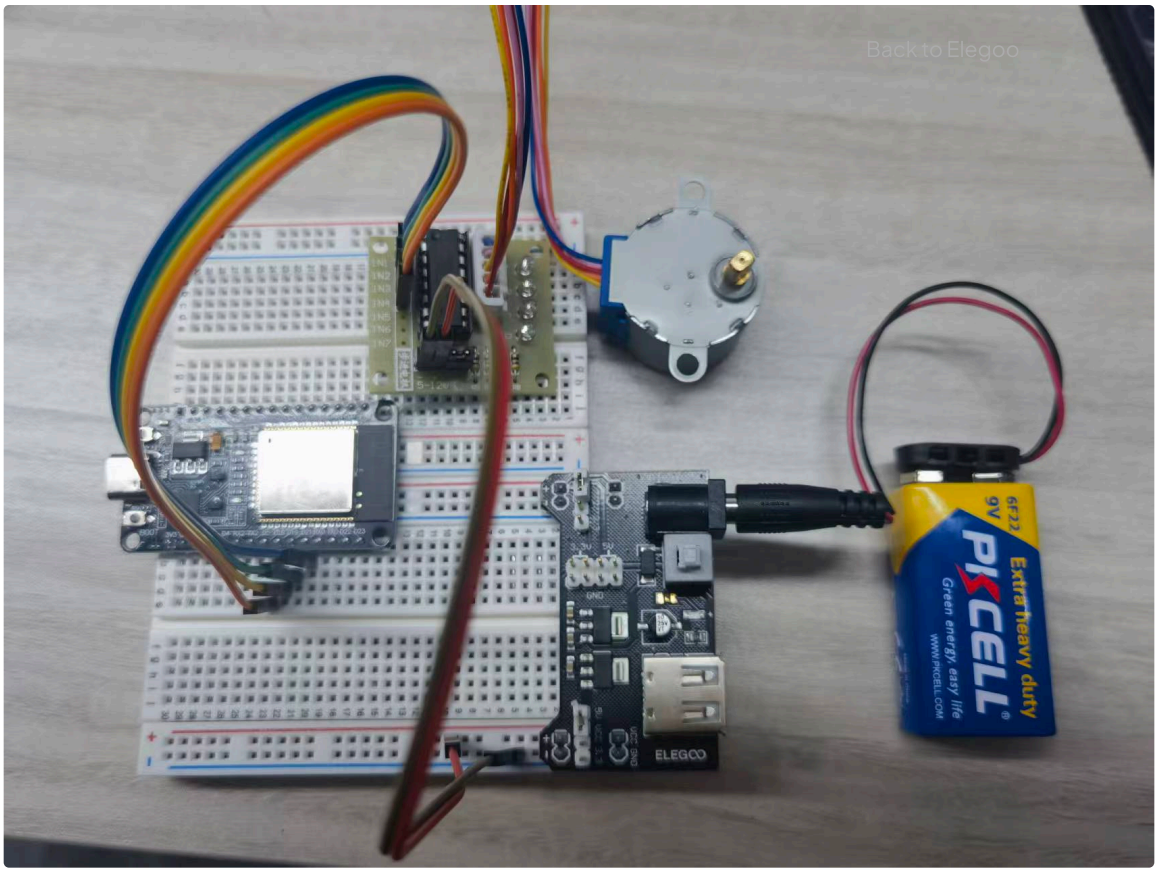
Connection Schematic



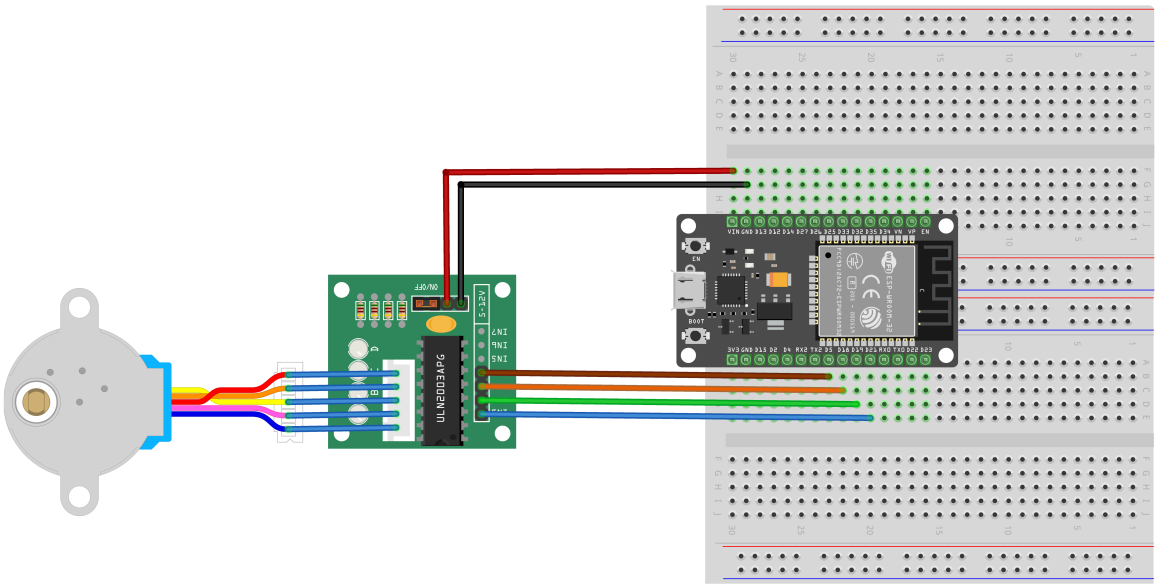
Wiring diagram



Practical Wiring Diagram



If you do not have a dedicated power supply, connect the positive terminal of the power source to the VIN pin of the ESP32 as shown in the figure below.



Code

You can click the blue text link to download the program file to your local device, and double-click the file to open it after the download is complete. Please note: Before opening the file, ensure that you have installed the Arduino IDE development environment and completed the installation of relevant components such as the board support package and driver corresponding to the ESP32 development board. If you have any questions about this operation process, you can refer to the "part 1" chapter of the document for detailed guidance.

DOWNLOAD stepper_motor.zip

With this code, the ESP32 board will control a 28BYJ-48 stepper motor to rotate clockwise and counterclockwise alternately, demonstrating basic stepper motor control using the Arduino Stepper library. The code also provides serial monitor feedback for each rotation direction.

Click the Serial Monitor button to turn on the serial monitor. The basics about the serial monitor are introduced in details in tutorial 4 in part 2.

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```
#include <Stepper.h>

const int stepsPerRevolution = 2048; // change this to fit the number of steps per revolution
const int rolePerMinute = 10; // Adjustable range of 28BYJ-48 stepper is 0~17 rpm

// initialize the stepper library on pins :
Stepper myStepper(stepsPerRevolution, 21,18,19,5);//In1-In3-In2-In4
```

Library and Hardware Configuration:

- `#include <Stepper.h>` : Includes the Arduino Stepper library for stepper motor control
- `const int stepsPerRevolution = 2048` : Defines the number of steps per full revolution for the 28BYJ-48 stepper motor
- `const int rolePerMinute = 10` : Defines the motor speed in revolutions per minute (RPM)
- `Stepper myStepper(stepsPerRevolution, 21,18,19,5)` : Creates a Stepper object with specified pins
  - Steps per revolution: 2048
  - Pin connections: IN1=21, IN3=18, IN2=19, IN4=5 (ULN2003 driver board pin order)

```
void setup(){
myStepper.setSpeed(rolePerMinute);
// initialize the serial port:
Serial.begin(9600);
}
```

setup() Function: Initializes the system and configures the stepper motor.

- `myStepper.setSpeed(rolePerMinute)` : Sets the motor speed in revolutions per minute (10 RPM)
- `Serial.begin(9600)` : Initializes serial communication at 9600 baud rate for operation feedback

Stepper Speed Setting:

- The setSpeed() function sets the approximate speed in RPM
- Actual speed may vary based on load and power supply
- 28BYJ-48 motors have a practical speed limit around 17 RPM

loop() Function Overview: The main loop that makes the stepper motor rotate alternately in clockwise and counterclockwise directions.

1. Clockwise Rotation Module:

```
// step one revolution in one direction:
Serial.println("clockwise");
myStepper.step(stepsPerRevolution);
delay(500);
```

- **Operation Description**: Makes the motor rotate one full revolution clockwise
- **Direction Indication**: Prints "clockwise" to the serial monitor
- **Step Control**: `myStepper.step(stepsPerRevolution)` moves the motor 2048 steps (one full revolution)
- **Delay**: 500ms pause after completion before next operation

2. Counterclockwise Rotation Module:

```
// step one revolution in the other direction:
Serial.println("counterclockwise");
myStepper.step(-stepsPerRevolution);
delay(500);
```

- **Operation Description**: Makes the motor rotate one full revolution counterclockwise

- **Direction Indication**: Prints "counterclockwise" to the serial monitor
- **Step Control**: `myStepper.step(-stepsPerRevolution)` moves the motor -2048 steps (one full revolution in reverse)
- **Delay**: 500ms pause after completion before next loop iteration

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Stepper Motor Operation:

- **Step Sequence**: The Stepper library uses a four-phase sequence (IN1→IN3→IN2→IN4) for half-stepping mode
- **Half-Stepping**: Provides smoother operation and double the resolution compared to full-stepping
- **Direction Control**: Positive step values for clockwise, negative for counterclockwise
- **Position Accuracy**: Stepper motors maintain their position without a feedback mechanism

Pin Configuration (ESP32 to ULN2003 Driver):

ESP32 Pin	ULN2003 Driver Pin	Motor Phase
GPIO 21	IN1	Phase 1
GPIO 18	IN3	Phase 3
GPIO 19	IN2	Phase 2
GPIO 5	IN4	Phase 4
5V	VCC	Power supply
GND	GND	Ground

Serial Monitor Output:

- "clockwise" - When the motor starts rotating clockwise
- "counterclockwise" - When the motor starts rotating counterclockwise

Troubleshooting:

- **Motor not moving**: Check power supply, connections, and pin configuration
- **Incorrect direction**: Swap any two adjacent phase wires
- **Rough operation**: Ensure proper step sequence and reduce speed
- **Stalling**: Reduce speed or check for excessive load

Stepper Library Functions:

- `Stepper(steps, pin1, pin2, pin3, pin4)` : Creates a 4-wire stepper motor object
- `setSpeed(rpm)` : Sets the motor speed in revolutions per minute
- `step(steps)` : Moves the motor a specified number of steps

←DC MOTOR Overview→

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